

Power System Implications of the New USB Power Delivery 3.2

Deric Waters

Distinguished Member of Technical Staff
at Texas Instruments

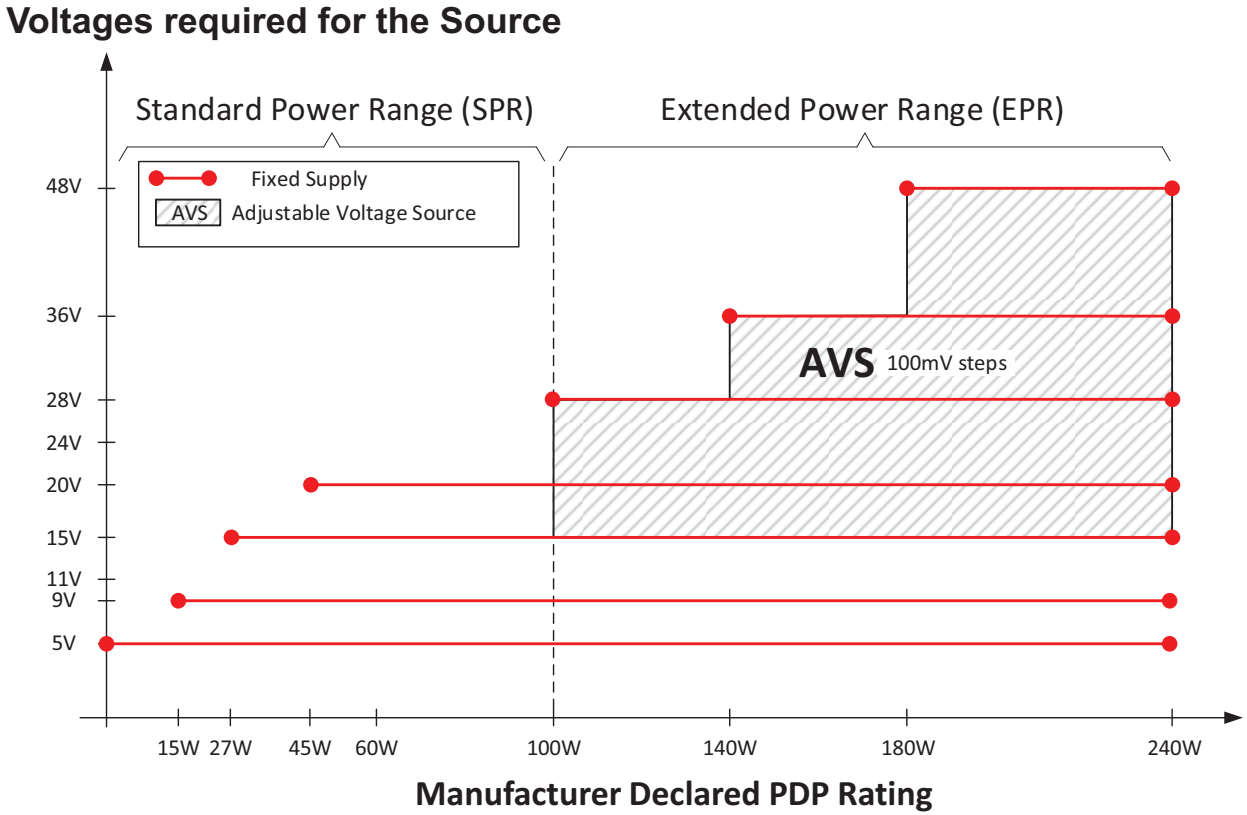
Texas Instruments

Agenda

- Major updates to the USB-C and USB-PD specifications.
  - Adjustable Voltage Source (AVS) requirements
  - Corrosion Mitigation
- Implications for source and sink systems

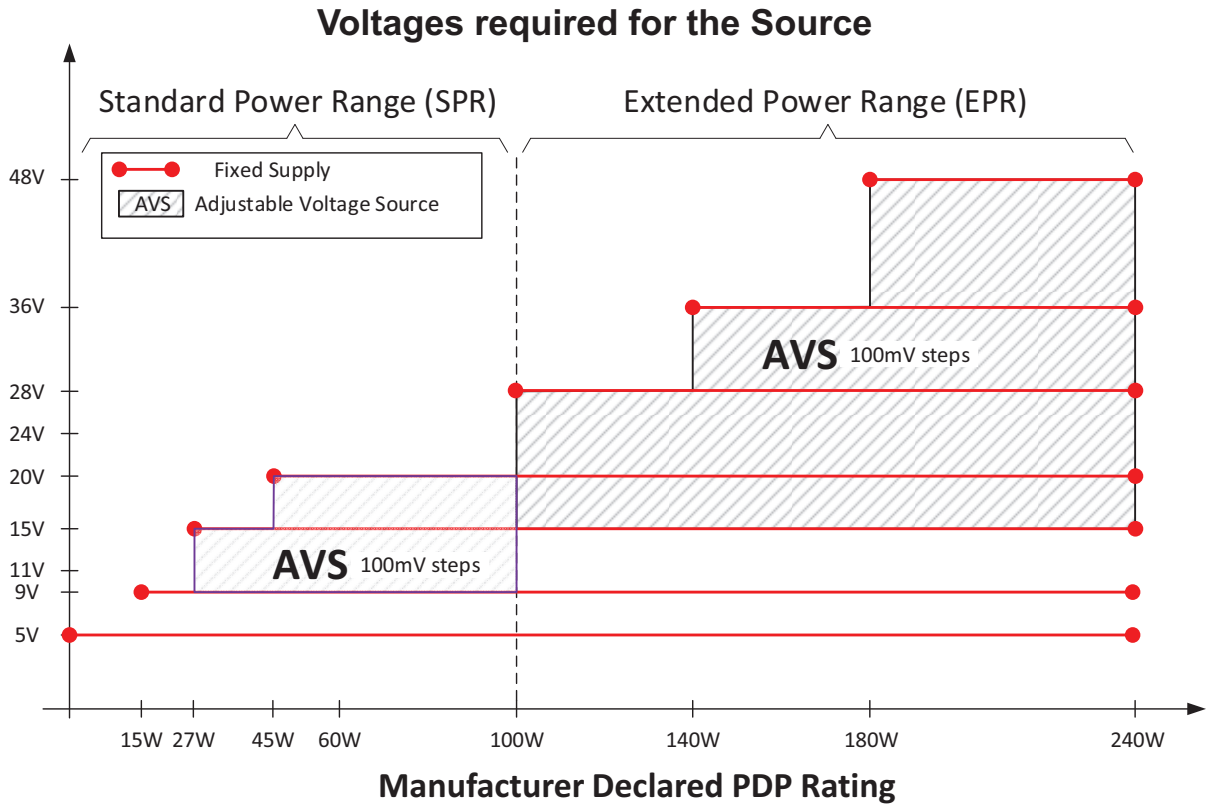
Summary of USB-PD Power Rules (R3.1)

- Chart shows which voltages a source with a given power rating must be capable to provide.
  - For <100W there were at most 4 voltages required.
  - PPS is optional and not shown.
  - Cable may limit to ≤ 20V
- Required current
  - ~PDP Rating ÷ Voltage
  - ≤ 5A
  - Cable may limit to ≤ 3A



Summary of USB-PD Power Rules (R3.2 as of October 2023)

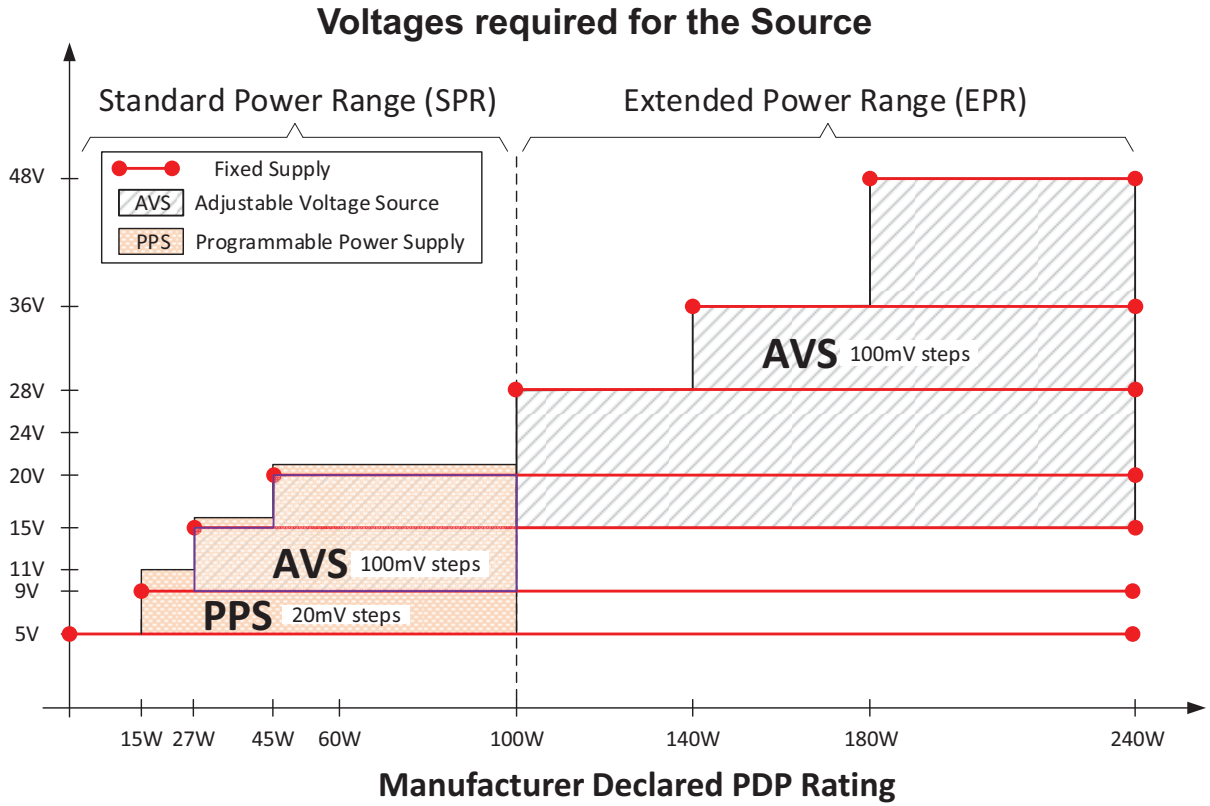
- Required current same as R3.1
- AVS now required for SPR
  - 27W < PDP ≤ 45W: 9V-15V
    - 59 more voltages than R3.1
  - 45W < PDP ≤ 100W: 9V-20V
    - 108 more voltages than R3.2
- Options allowed
  - Extra current
  - Extra voltages <9V only,
  - Additional restrictions for EPR
    - No extra voltages at all



Summary of USB-PD Power Rules (R3.2 as of October 2023)

- Required current same as R3.1
- AVS now required for SPR
  - 27W < PDP ≤ 45W: 9V-15V
  - 45W < PDP ≤ 100W: 9V-20V

- Options allowed
  - Extra current
  - Extra voltages <9V only,
  - Additional restrictions for EPR
    - No extra voltages at all
    - No Variable supply while in EPR mode
  - PPS (with 5V minimum)
    - Increased from 3.3V



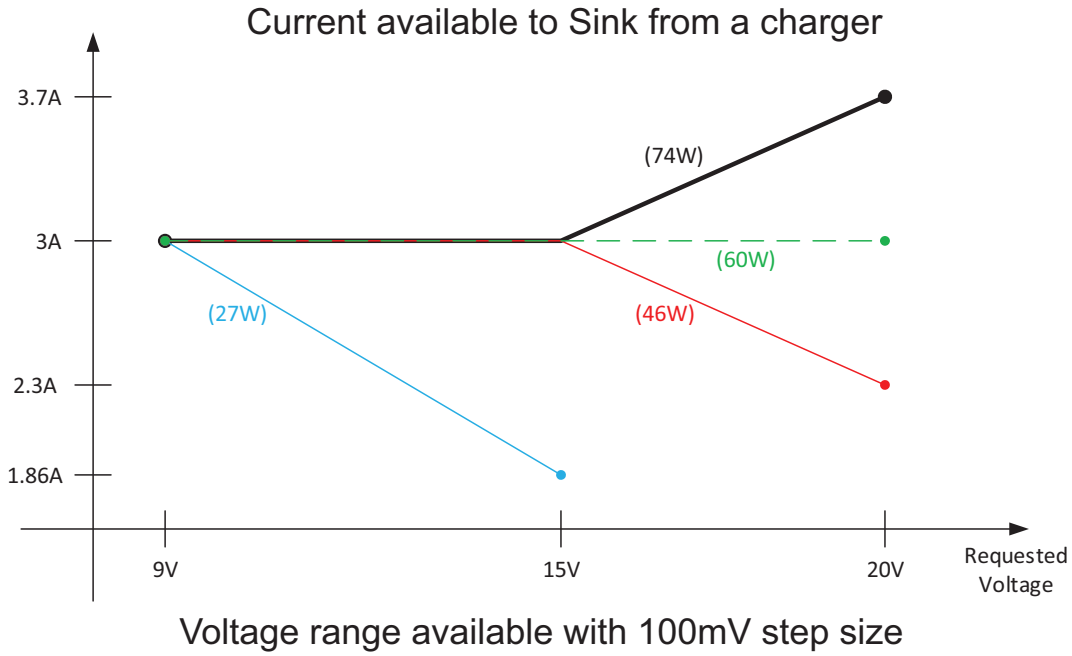
AVS ≠ PPS

PPS = Programmable Power Supply
AVS = Adjustable Voltage Supply

- AVS and PPS both have programmable voltages (different step sizes).
- Only PPS requires a current limit (50mA steps, +/-5%)
  - Relaxed from -25mA/+100mA to (±150mA or ±5%).
- A USB-IF certified charger may or may not include PPS.
  - USB-IF certifies “Fast Charger” with PPS
  - USB-IF certifies “Charger” without PPS
- A “Fast Charger” implements PPS & can “easily” add AVS.
- AVS is a big change for “Charger” implementations.
  - Because they don’t already implement PPS
- Some sinking products will charge faster if PPS is available. It is a cost trade-off for the source.

Expanding AVS below 100W

- Why add AVS?
  - Avoid one-off chargers with a specific/unique voltage → achieve universal behavior.
  - A sink can be designed for a specific voltage since USB PD 3.2 chargers provide that voltage.
    - Sinks should still be designed to function when connected to an older adaptor.
  - Ex. Sinks can improve charging efficiency by selecting an input voltage closer to the battery voltage when connected to PD 3.2 charger.



Types of DC/DC for sinking applications

Application needs	Functionality when desired voltage unavailable	Type of DC/DC Needed
Sink requires ≤4.0V* (eg. 1S battery-powered device)	N/A	DC/DC buck
Sink requires >4.0V*	Non-functional	DC/DC buck
	At least some functionality required (eg. Slow charge)	DC/DC buck/boost

* USB-C specified tolerance allows 4.75V at the source end of the cable.
USB-C allows 0.75V of IR drop when VBUS current is 3A.
Therefore a sink may see as low as 4.0V, when nominal VBUS=5V.

Types of DC/DC for sourcing applications

Input Voltage		DC/DC Needed
DC input is higher than required VBUS	Buck performance across the voltage range acceptable.	DC/DC buck
	Buck/Boost preferred for better thermal performance and efficiency	DC/DC buck/boost
DC input not always high enough	N/A	

Poor buck efficiency from 48V to 5V is likely worse than boost efficiency from 30V to 48V. So using an input DC voltage of ~30V can be better.

What is required VBUS?

- Max VBUS_nom = max nominal VBUS
  - This may be 5V, 9V, 15V, 20V, 28V, 36V or 48V depending on the power rating.
- USB-PD allows -5% for tolerance, and allows 0.5V transient droop for load steps.
- Ex. VBUS_nom=20V → VBUS > 18.5V required (including transients)

What if the source allow VBUS to droop below this for a short time?

- USB-C *requires* sinks to disconnect if VBUS droops more than expected.
  - This can lead to poor user experience (ex. Audible beeps from a phone)
    - Sinks typically disconnect without much or any debouncing to mitigate arcing, so even short transients may cause a disconnect.
- Example for VBUS_nom=20V:
  - If the sink sees VBUS < 17.75V it *may* disconnect, if < 15.975V it *shall* disconnect.
  - Remember 0.75V is reserved for cable IR drop 17.75V + 0.75V = 18.5V

Multi-port chargers

- Chargers have the same normative requirements for each port.
- Assured-Capacity multi-port charger
  - Always offers the same power on its ports regardless of what is connected.
- Shared-Capacity multi-port charger.
  - The PD controller has the capability to force the sink to reduce its requested power.
  - In case the power is no longer available (another port needs it, or to reduce thermals)

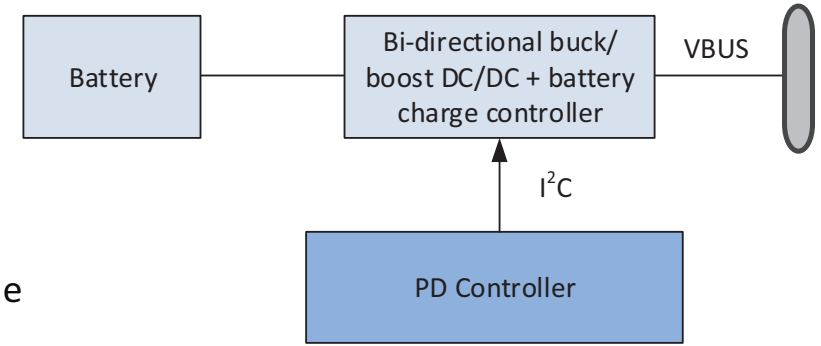
What the sink *needs*		What the sink *gets*			
Port 1	**Port 2**	**Port 1**	**Port 2**	**Port 1**	**Port 2**
0 (unconnected)	60W	0	30W	0	60W
60W	60W	30W	30W	7.5W-52.5W*	7.5W-52.5W*
15W	45W	30W	30W	15W	45W
		Assured-Capacity Charger		Shared-Capacity Charger	

* The total of both ports ≤ 60W. The sharing algorithm is up to implementers. At least 7.5W must be given to a port.

What if a source offers insufficient power?

- Shared capacity chargers are becoming more popular, since much of the time not all ports on a charger are in use. So ports in use can offer more power for better user experience.
- Due to thermal conditions (ex. In automotive USB-C charging) even an Assured Capacity charger may reduce the available power at times.

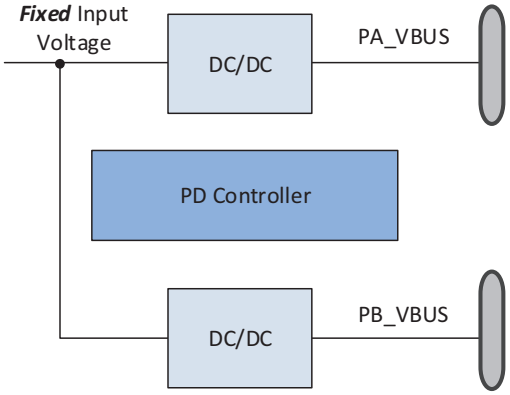
- A sink must handle being forced to reduce power
  - A buck/boost battery charger topology is a good solution. Charge the battery from 5V by boosting up to battery voltage. Or charge more efficiently when VBUS is higher using the buck mode.
  - If the reduced power is lower than desired assert the Capability Mismatch bit.
    - The USB-PD spec requires sources that have reserve power to increase the amount of power offered to the sink within 2 seconds.



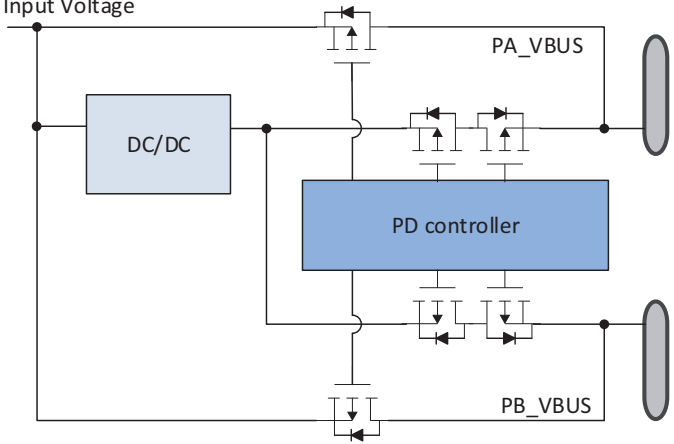
2-port AC/DC comparison (DC/DC arch.)

Arch	# key components	Voltage limitations	Current limitations
#1	2 DC/DC buck: 4 FETs, 2 Ind.	No port dependencies (Best user experience). Input voltage need not be adjustable.	No port dependencies (Best user experience).
#2	1 DC/DC buck: 8 FETs, 1 Ind.	For one port, PD controller sets input voltage. Other port's voltage cannot exceed the input voltage. Complex FET sequencing.	There cannot be an accurate current limit on the input voltage … PPS is not possible on both ports.
#3	1 DC/DC buck: 5 FETs, 1 Ind.	For Port A, PD controller sets the input voltage. PortB's power sometimes limited by boost efficiency.	There cannot be an accurate current limit on the input voltage … PPS is not possible on both ports.

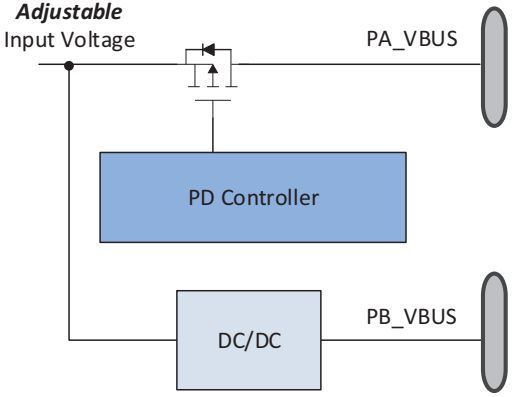
Architecture #1



Architecture #2



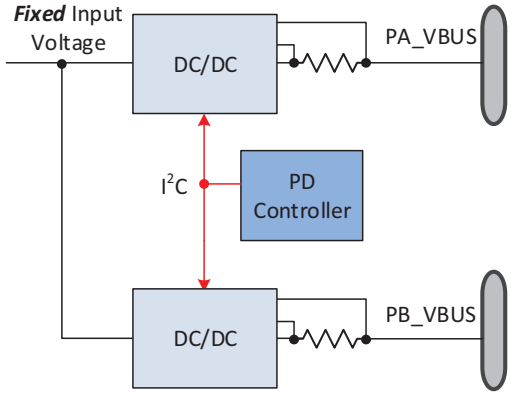
Architecture #3

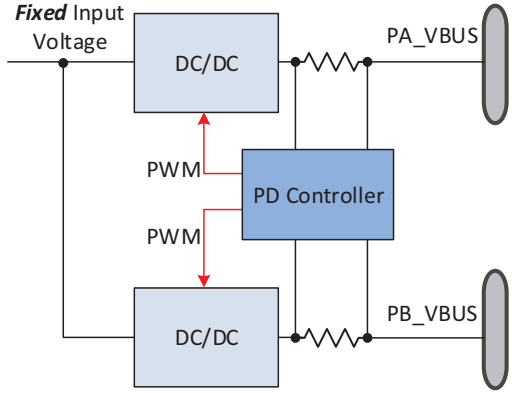


2-port AC/DC comparison (DC/DC Control)

Feature	I2C Controlled	PWM Controlled
PPS capable	Yes, if DC/DC has configurable ILIM.	Yes, but requires PD controller to sense current and regulate the PWM.
Monitoring	Comprehensive: input/output OVP, OCP, UVP, thermal	Limited to simple monitoring via GPIO.
Configurable features	Adjust modes (eg. light load efficiency.) Configure EMI suppression modes	N/A
Current sensing	Can be done by the DC/DC. Not needed in PD controller.	Must be in the PD controller for PPS.
Noise sensitivity	Internal DAC less sensitive	Manipulating REF voltage more sensitive
# of PD controller pins	Only 2 pins for any # of ports (config address for DC/DC)	1 pin per port

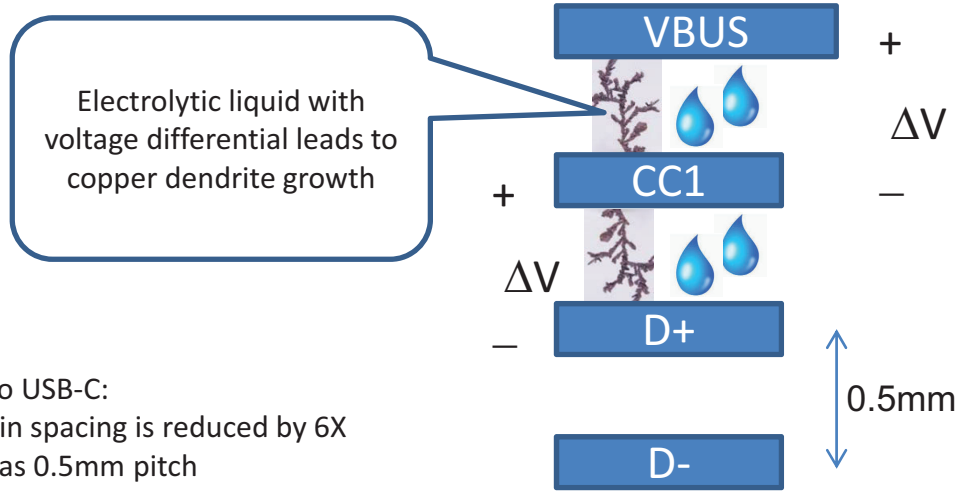
Showing DC/DC architecture #1, but the control can apply to any DC/DC architecture.





Corrosion caused by liquid in the connector

- Dipping a cable plug in & out of a liquid, then plugging the cable plug into a receptacle can damage the receptacle.
  - Example at right with liquid = Powerade.

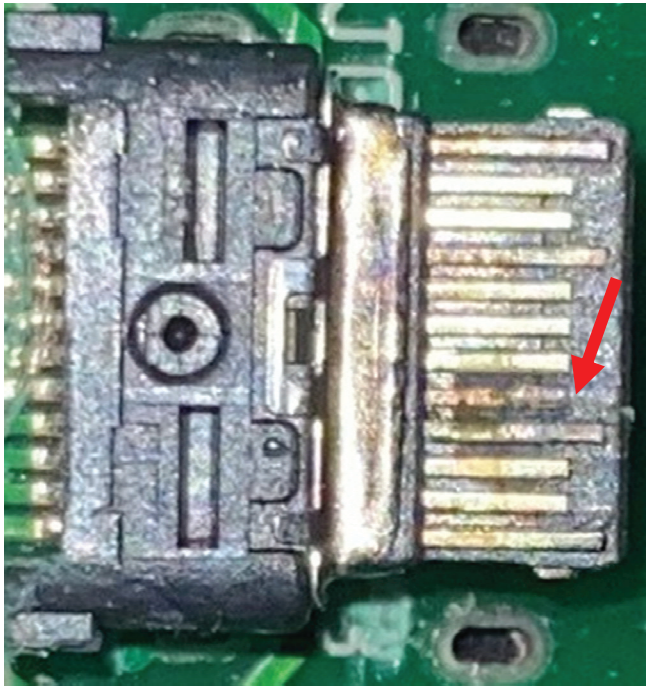


USB-A 2.0 to USB-C:

- Pin-to-pin spacing is reduced by 6X
- USB-C has 0.5mm pitch

- The USB-C spec has added guidance for detecting and mitigating damage due to liquid.

After removing outer shell from receptacle.



Conclusion

- Come see our USB-PD demo at the TI booth